Characterization of Silicon Strip Sensors

Leonardo Spallacci, Woradon Sophonamphonsucha

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Abstract

This laboratory experiment investigates the operational principles and physical properties of a silicon strip semiconductor detector, utilizing technology analogous to that employed in the ATLAS experiment at the Large Hadron Collider (LHC). Using the Educational Alibava System (EASY), we characterized a 300 μm *thick p-in-n silicon sensor behavior by determining its depletion voltage, evaluating intrinsic electronic noise (pedestals and common mode shifts), and performing a charge calibration of the analog-to-digital converter (ADC). Furthermore, the spatial configuration and the Charge Collection Efficiency (CCE) of the sensor as a function of the bias voltage were examined using an infrared laser, yielding measurements of the strip pitch and laser penetration depth, confirming the full depletion regime of the device.*

1 Introduction

In modern high-energy physics, resolving the trajectories of charged particles with extreme precision is fundamental to understanding fundamental interactions. At the European Organization for Nuclear Research (CERN), the Large Hadron Collider (LHC) collides protons at unprecedented center-of-mass energies (up to $\sqrt{s} = 13.6\,\text{TeV}$) to test the predictions of the Standard Model and probe for new physics beyond it [1].

To effectively analyze the highly dense and complex collision environments produced by the LHC, experiments require robust, fast, and highly granular tracking systems. Silicon strip detectors have become the standard technology to fulfill these requirements. The primary motivation for this laboratory exercise is to investigate the physical principles underlying these devices, specifically focusing on the technology utilized in the ATLAS experiment [2].

By characterizing a simplified $300\,\mu$m thick p-in-n silicon strip sensor using the Educational Alibava System (EASY) [3], we aim to determine important parameters such as the depletion voltage, intrinsic electronic noise, signal calibration, and Charge Collection Efficiency (CCE). To provide a comprehensive background for these measurements, this section will first outline the role of silicon trackers in the ATLAS detector, followed by explaining the semiconductor physics that governs the sensor's operation, and finally present the experimental setup utilized for data acquisition.

1.1 The ATLAS Detector

One of the primary experiments situated on the LHC ring is the ATLAS detector, a massive, multi-layered cylindrical apparatus [2] (Figure 1). From the inside out, the detector is composed of three main systems: the Inner Detector (ID) for tracking, the electromagnetic and hadronic calorimeters for energy measurement, and the outermost muon spectrometer for identifying and measuring the momenta of muons. These sub-detectors are arranged concentrically around the beam axis, providing almost full solid angle coverage. A strong 2 T magnetic field, generated by a superconducting solenoid surrounding the ID, bends the trajectories of charged particles, allowing for the determination of their momentum and charge.

This experience focuses on the technology used in the Inner Detector (ID), which is the innermost part of ATLAS, located just a few centimeters from the proton-proton collision point. The ID (shown in Figure 2) is designed to provide highly precise tracking and vertexing information. It is structured in three concentric layers. First, the Pixel Detector: it's the innermost layer, consisting of highly segmented silicon pixels. It provides extremely high spatial resolution (almost $10\,\mu$m) necessary to reconstruct primary and secondary interaction vertices close to the beam pipe. Second, the Semiconductor Tracker (SCT): surrounding the pixel detector, the SCT is composed of four double layers silicon strip sensors, similar to the one characterized in this laboratory experiment. By requiring particles to traverse at least four layers of silicon, the SCT achieves a

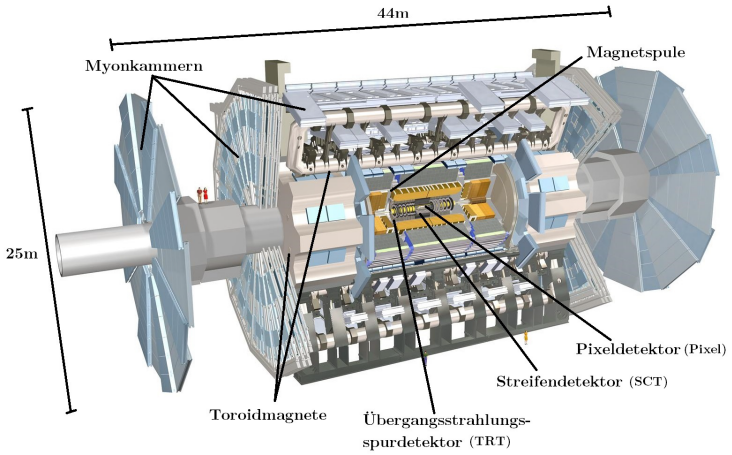


Figure 1: Schematic view of the ATLAS cylidrical spectrometer.

spatial resolution of roughly 25 μm, which is crucial for precisely reconstructing the tracks of charged particles over a larger volume. Lastly, the Transition Radiation Tracker (TRT): it's the outermost part of the ID, made of thousands of thin, gas-filled drift tubes. It provides continuous tracking with a lower spatial resolution but a high number of hits per track, helping in electron identification.

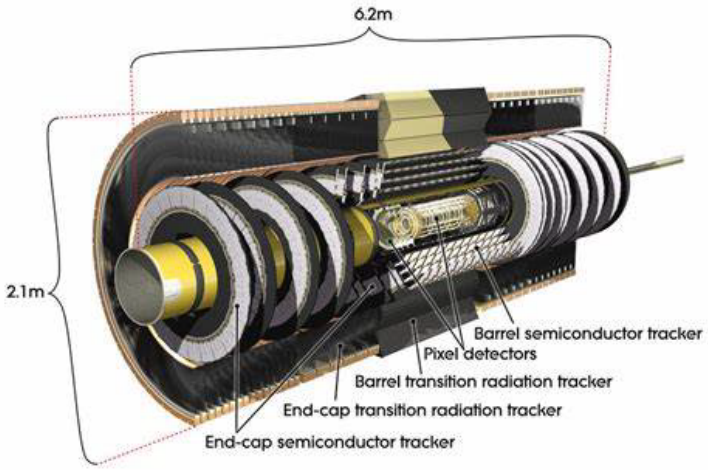


Figure 2: Schematic representation of the ATLAS inner detector.

1.2 Silicon Strip Sensors

Semiconductors, such as silicon, have electrical conductivity properties that fall between those of conductors and insulators [4] [5]. Their electronic structure is characterized by a valence band and a conduction band empty, which are separated by a band gap of moderate energy ($E_g \approx 1.12\,\text{eV}$ for silicon at room temperature). The silicon used in detectors is typically modified through doping, the intentional introduction of impurity atoms into the crystal lattice (Figure 3). The doping can be of two types: n-type, when elements from group V (e.g., Phosphorus) are used to introduce excess electrons; and p-type, when elements from group III (e.g., Boron) are used to introduce "holes". Holes are basically vacancies of electrons that act as mobile positive charges. This carrier generation occurs because silicon atoms possess four valence electrons; introducing a pentavalent atom leaves one extra electron free to conduct, whereas a trivalent atom lacks one electron to complete the local covalent bonds, thus creating a mobile vacancy.

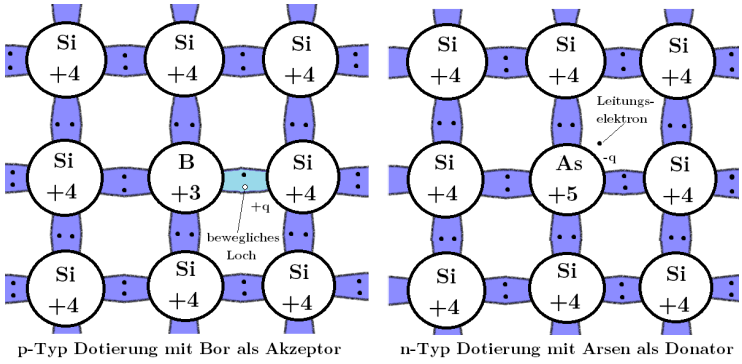


Figure 3: Schematic representation of an n- and p-type doping in a two-dimensional silicon lattice.

The fundamental core of the detector studied in this experience is a p-n junction. It is formed at the interface between two semiconductors, a p-type layer and an n-type layer. At this interface, the majority charge carriers (electrons from the n-side and holes from the p-side) diffuse and recombine, creating a region without free charge carriers called the depletion zone. To operate the device as a detector, a reverse bias is applied (positive voltage to the n-side and negative voltage to the p-side). This process widens the depletion zone until it occupies the entire thickness of the sensor ($D = 300\,\mu\text{m}$). The voltage required to reach this state is known as the depletion voltage (U_{dep}).

In this configuration, the depletion zone acts like a solid-state ionization chamber:

1. In the absence of external radiation, no significant current flows (minimal leakage current) since there are no free charge carriers.

2. When the laser photons or an ionizing particle traverses the depletion region, it deposits energy, creating electron-hole pairs.

3. The internal electric field immediately separates these charges, drifting them towards the electrodes (the strips), thereby generating the electrical signal that is subsequently digitized into ADC counts by a readout chip.

1.3 Experimental Setup

The experimental apparatus used for this laboratory exercise is the Educational Alibava System (EASY). It is a compact and simplified version of a high-energy physics data acquisition system, specifically designed to characterize silicon micro-strip sensors. The system is composed of three main components: the Detector Unit, the Control Unit, and a computer running the Alibava-gui software for data acquisition and system control.

Detector Unit

The detector unit houses the readout electronics and the sensing element: a 300 µm thick, n-doped silicon bulk with 128 p-doped parallel implants (Figure 4). This p-in-n sensor design is analogous to those used in the ATLAS SCT. A continuous bottom metallization layer provides the reverse bias voltage contact (up to 200 V). To AC-couple the signals and block leakage current from the sensitive electronics, aluminum readout strips are deposited over the p-implants, separated by an insulating SiO2 layer. These 128 strips are wire-bonded to a BEETLE readout chip, which amplifies and temporarily stores the radiation-induced charge signals before transmitting them to the Control Unit for ADC digitization. Above the sensor, a sliding mechanical platform hosts an integrated infrared laser system used for spatial characterization and Charge Collection Efficiency (CCE) measurements. The laser (980 nm wavelength, 5 ns pulse length, 0.6 mW peak power [3]) is delivered via a fiber optic cable. Two micrometer screws provide vertical focusing and horizontal positioning across the strips with a 10 µm accuracy, allowing precise signal generation mapping across the detector without the need for a physical radioactive source.

Control Unit

The control unit serves as the central interface for the EASY system, supplying power and managing data transmission between the detector unit and the local computer. Crucially for the experimental measurements, it provides the variable reverse bias voltage to the sensor and monitors the corresponding leakage current with a resolution of 0.01 μA, which is fundamental for the I-V characterization. Additionally, the unit generates the infrared laser pulses and manages the trigger signals required to synchronize the data acquisition during the spatial and Charge Collection Efficiency (CCE) measurements. Care was taken throughout the experiment to ensure that the sensor was strictly operated in a reverse-bias regime to prevent conduction.

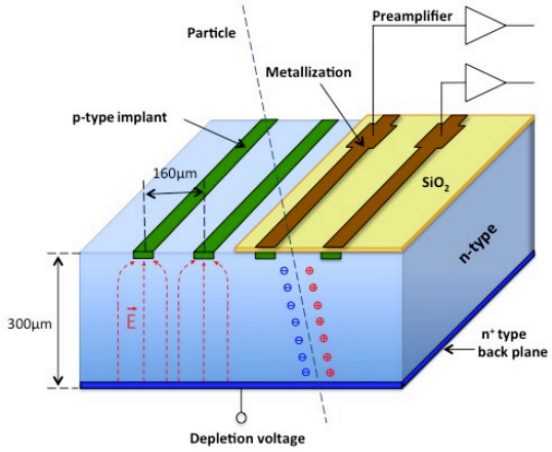


Figure 4: Schematic cross-section of the p-in-n silicon micro-strip sensor, illustrating the p-implants, the oxide layer, the AC-coupling metallization, and the particle interaction.

This whole system is managed by the user on the local computer thanks to Alibava software, and its Graphic User Interface (GUI). The Alibava software's functioning is explained in more detail in [3].

2 Analysis

2.1 Current-Voltage Characteristic

To efficiently measure the signals generated by passing particles, a semiconductor detector must be fully depleted. To determine the precise voltage required to reach this state, an I-V (Current-Voltage) characteristic measurement was carried out. This is necessary because, at low voltages, the leakage current rises rapidly with the applied bias due to the presence of free charge carriers in the sensor bulk. However, once full depletion is reached, the rate of current increase slows down significantly. We thus measured the I-V curve from 0 to 200 V in 10 V steps to determine the actual depletion voltage (U_{dep}). The resulting characteristic is shown in Figure 5.

As observed in the plot, the current increase begins to flatten at approximately 60 V. Thus, we identified this value as U_{dep}, which is consistent with the manufacturer's specifications [3]. To ensure complete depletion of the sensor for all subsequent measurements, we chose to apply an operating voltage at least 20 V higher than U_{dep}, fixing the bias at 88 V.

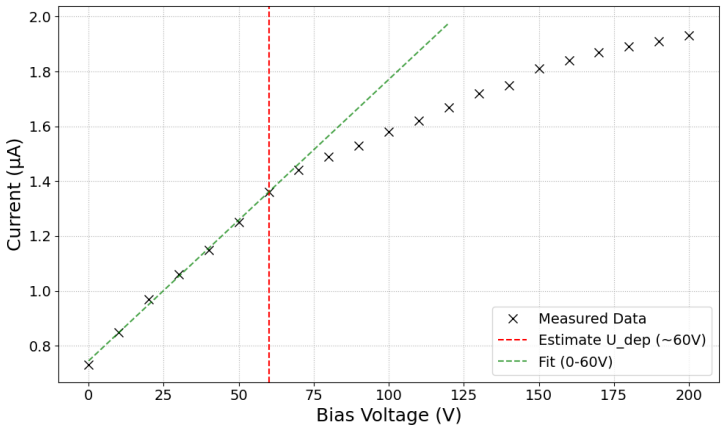


Figure 5: Current-voltage characteristic of the sensor. A fit in the $0 - 60\,\mathrm{V}$ interval is shown in green, before the linear increase due to reaching the depletion regime. The red line indicates the chosen value for U_{dep}.

2.2 Measurement of Pedestals and Noise

The sensor itself and the readout electronics generate noise signals that interfere with the actual relevant signals. So in order to minimize this noise[1] it's fundamental to be able to characterize it. The total raw signal, expressed in ADC counts, recorded by a generic strip i during an event k can be modeled as the sum of three distinct components:

$$\mathrm{ADC}(i,k) = P(i) + D(k) + \mathrm{Signal}(i,k) \tag{1}$$

Here, the pedestal $P(i)$ represents the intrinsic baseline offset of the i-th strip in the absence of any external physical signal. It is calculated as the mean value of the ADC counts over N consecutive measurements taken without a radiation source:

$$P(i) = \frac{1}{N}\sum_{k=1}^{N} \mathrm{ADC}(i,k) \tag{2}$$

The Common Mode Shift $D(k)$ accounts for global electrical fluctuations that simultaneously affect all 128 strips during a single event k. It is computed by averaging the pedestal-subtracted signals across all channels:

$$D(k) = \frac{1}{128}\sum_{i=1}^{128} \left(\mathrm{ADC}(i,k) - P(i)\right) \tag{3}$$

Finally, the intrinsic noise of each strip is evaluated as the Root Mean Square (RMS) of the ADC counts, after systematically subtracting both the individual

[1] Noise is common to all types of experiments, especially in particle physics. It's basically impossible to fully eliminate it, so usually physicists aim to reduce it as much as possible.

pedestal and the event-specific common mode shift:

$$\text{Noise}(i) = \sqrt{\frac{1}{N-1}\sum_{k=1}^{N}\left(\text{ADC}(i,k) - P(i) - D(k)\right)^2} \tag{4}$$

To measure these quantities experimentally, a dedicated *Pedestal Run* consisting of $N = 1000$ events was performed using the Alibava software, keeping the setup shielded and without any external radiation source. The recorded raw data was then processed via Python scripts to map the pedestals, the common mode distribution, and the noise profile for all the channels. The results of the pedestal and noise calculations for the 128 channels are shown in Figure 6.

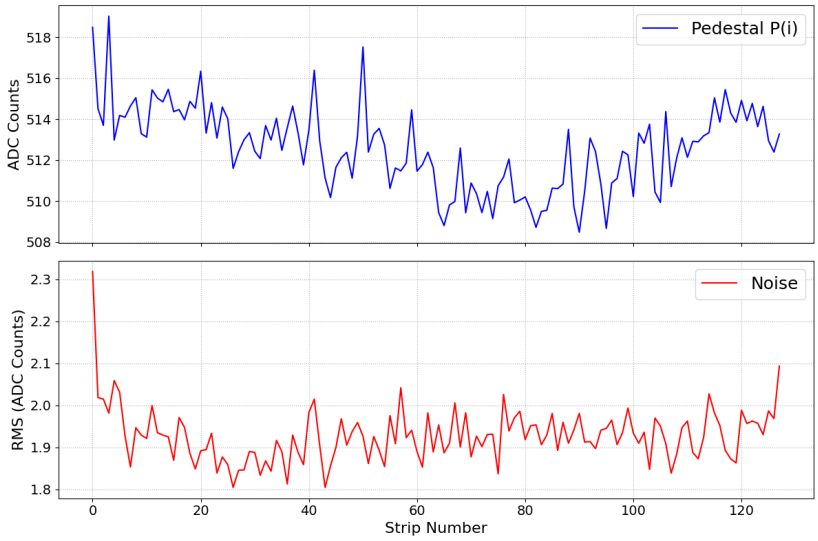


Figure 6: Overview of the pedestal (top, blue) and intrinsic noise (bottom, red) for each of the 128 strips of the sensor.

The upper plot displays the pedestal values, which fluctuate between roughly 508 and 519 ADC counts. The lower plot illustrates the intrinsic noise of each individual strip. The overall noise level is extremely low and stable, averaging around 1.9 ADC counts (RMS). The stability of this value across the detector indicates that all strips are well-bonded and fully functional. A minor noise spike is visible at strip 0 (reaching $\sim$ 2.3 ADC counts), which is a common edge-effect phenomenon, likely due to its proximity to the bias ring or guard ring structure, leading to a slightly higher parasite capacity.

Finally, the Common Mode Shift distribution is presented in Figure 7. As anticipated by the theoretical models, the global disturbance affecting the events follows a distinct Gaussian distribution centered exactly at zero. This confirms that the common mode is a purely random electronic fluctuation with no underlying systematic bias.

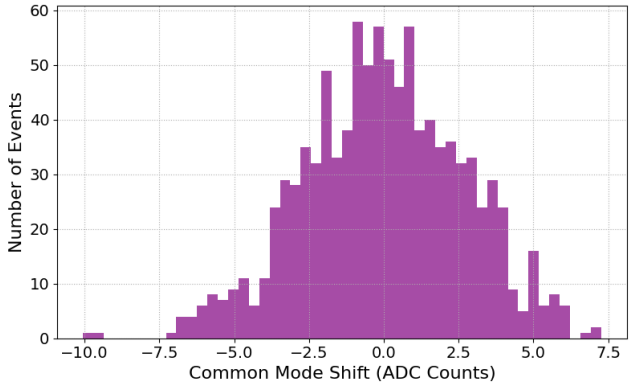


Figure 7: Distribution of the Common Mode Shift calculated over 1000 events.

2.3 Calibration

During the calibration procedure, defined electronic pulses are injected from the control unit into the BEETLE readout chip, which then outputs the corresponding signals in ADC counts. To correlate these digital values with the actual physical charge deposited in the sensor, it is necessary to establish a calibration curve. This conversion is crucial because it allows us to express the ADC counts in terms of the number of electrons, and consequently, the deposited energy (knowing that 3.6 eV are required to generate an electron-hole pair in silicon).

Before performing the charge calibration, the synchronization between the injected signal and the BEETLE chip readout must be optimized. This was achieved by executing a *Delay Scan*. The proper delay is defined by the time at which the sampled pulse reaches its maximum amplitude. As shown in Figure 8, the optimum delay was found to be 65 ns, which was subsequently set as the default parameter for the *Calibration Run*.

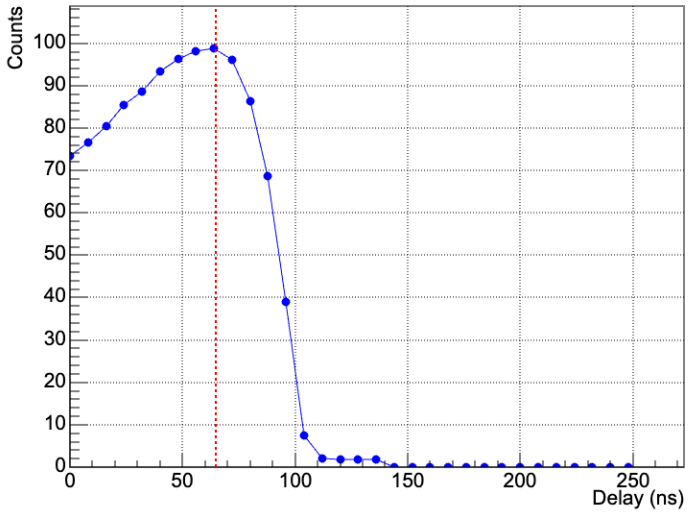
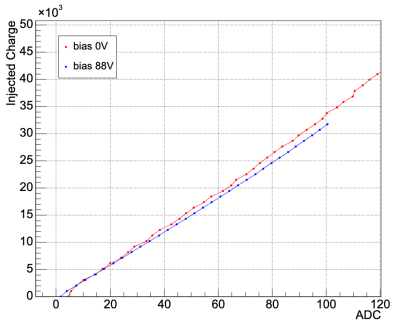
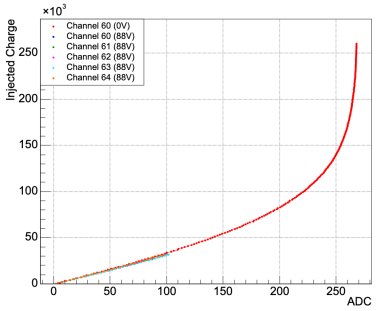


Figure 8: Delay measurement for calibration. The dashed line indicates the delay at which the signal reaches its maximum peak.

Following the delay determination, a charge calibration scan was performed for five adjacent channels (from 60 to 64). During this phase, the sensor was set at 88 V, ensuring it was operated well above the depletion voltage determined in Section 2.1. Additionally, a reference measurement at a 0 V bias was recorded for channel 60 to observe the system's response without full depletion.



(a) Calibration curves for the mean value of five selected channels (60-64) with bias voltage of 88V, and for channel 60 with no bias voltage (in the ADC range [0-110]).



(b) Calibration curves for the five selected channels (60-64) with bias voltage of 88V, and for channel 60 with no bias voltage (in the full ADC range).

Figure 9: Comparison curves in the full ADC range (right) and with a zoom in the [0-110] ADC range (left). This shows the missing data for the measurements with bias voltage of 88V.

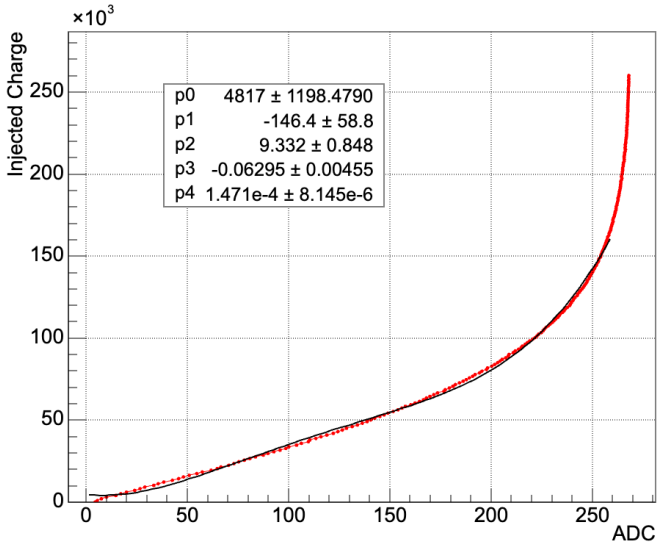


Figure 10: Calibration curve with bias voltage of 0 V with the polynomial fitting in the [0-260] ADC range to prevent overfitting (black line).

As observed in Figure 9b, the response curves for the five adjacent channels are nearly identical, demonstrating a uniform behavior of the readout electronics across this section of the chip. The conversion between ADC counts and the injected charge was obtained by fitting the calibration data with a 4th-degree polynomial function: $f(x) = p_0 + p_1 x + p_2 x^2 + p_3 x^3 + p_4 x^4$ in the specific range to prevent the overfitting. The resulting fitting parameters are summarized in Table 1. We can also see that the datasets from the five channels with large bias voltage are smaller than the dataset with 0V (Figure 9a). This missing data is due to the fact that by mistake we haven't increased the number of injected charges in the readout system from the software settings, before starting the calibration run.

Parameter	Value with uncertainty
p_0	$(4.82 \pm 1.19) \times 10^3$
p_1	$(-1.46 \pm 0.58) \times 10^2$
p_2	$(9.33 \pm 0.85) \times 10^1$
p_3	$(-6.29 \pm 0.45) \times 10^{-2}$
p_4	$(1.47 \pm 0.08) \times 10^{-4}$

Table 1: Fitting parameters of the calibration curve.

2.4 Spatial Configuration of the Sensor

For this measurement, the infrared laser was utilized to investigate the spatial structure of the silicon strip sensor, specifically to determine the strip pitch and the extension of the laser spot size.

Before performing the spatial scan, the system had to be synchronized. This was achieved by optimizing the delay between the trigger pulse sent by the Control Unit and the actual signal acquisition, ensuring that the system samples exactly at the peak of the laser-generated signal. As shown in Figure 11, the optimal delay was determined to be 100 ns. Subsequently, the laser was carefully focused and positioned using the vertical and horizontal micrometer screws to maximize the signal amplitude at a specific location on the detector.

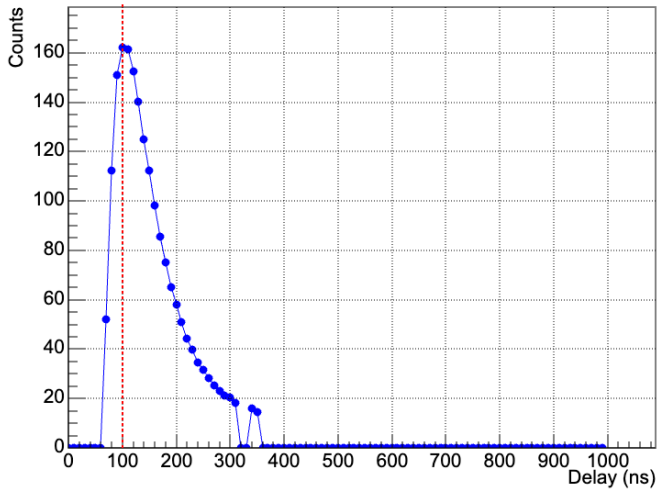


Figure 11: Delay measurement for laser synchronization. The dashed line indicates the optimal delay for this measurement (100 ns).

Once synchronized, the structural measurement was performed by translating the laser horizontally in 10 μm intervals across 35 distinct points, recording 1000 events at each step. The sensor's signal response as a function of the laser position is illustrated in Figure 12.

According to the plot in Figure 12, the signal intensity varies dramatically depending on whether the beam strikes the silicon bulk or the aluminum strips. When the laser is positioned directly above the metallization of strip 77, the beam is entirely reflected, causing the measured ADC counts to drop to a minimum of near-zero. Instead, when the laser is positioned in the silicon gap immediately to the left or right of the metal strip, it penetrates the bulk, generating a maximum signal peak. As the laser continues to move away from strip 77 and approaches an adjacent strip (e.g., strip 76 or 78), the signal on

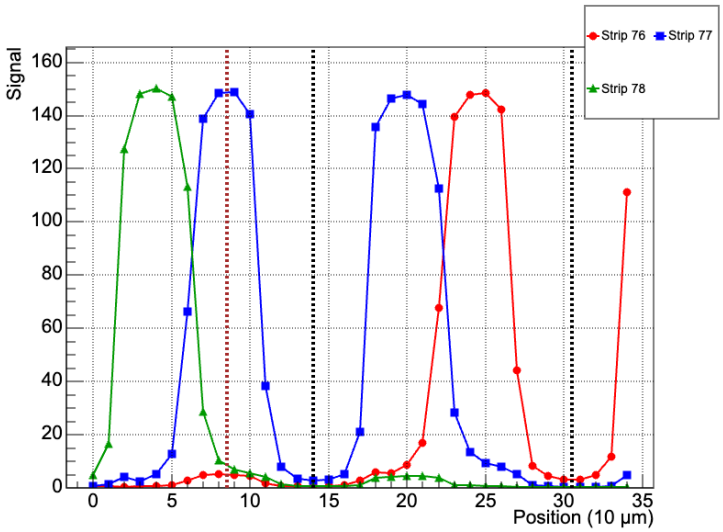


Figure 12: Channel signal amplitude as a function of the horizontal laser position. The black dashed line indicates the minimum, and the brown dashed line indicates the maximum peak of strip 77.

strip 77 decreases while the signal on the neighboring strip simultaneously rises, repeating this pattern across the whole sensor.

By analyzing these geometric signal fluctuations, the physical dimensions of the sensor can be extrapolated. The distance between two consecutive signal minima corresponds to the distance between the centers of two adjacent metallic bands, which defines the strip pitch. By evaluating the distance between the two minima in the plot, the pitch is estimated to be approximately 160.0 μm. Furthermore, the spatial extension of the laser spot can be deduced by measuring the distance between a signal maximum (laser fully in the gap) and the adjacent minimum of the same strip (laser fully blocked by the metal). From our data, the laser spot size is estimated to be 60.0 μm.

2.5 Measure of the Charge Collection Efficiency

A measurement of the Charge Collection Efficiency (CCE) of the sensor was performed using the infrared laser. The bias voltage was varied from 0 V to 200 V in steps of 10 V, recording 1000 events for each voltage step. For each measurement, the maximum signal amplitude (the peak ADC count of the strip hit by the laser) was extracted. These values were then normalized to the absolute maximum observed signal to express the efficiency on a scale from 0 to 1. As illustrated in Figure 13, the CCE increases steeply at voltages below $U_{dep} \approx 60$ V. Once the bias exceeds the depletion voltage, the efficiency flattens and remains constant at a value close to 1, confirming that the active volume

of the sensor is fully depleted.

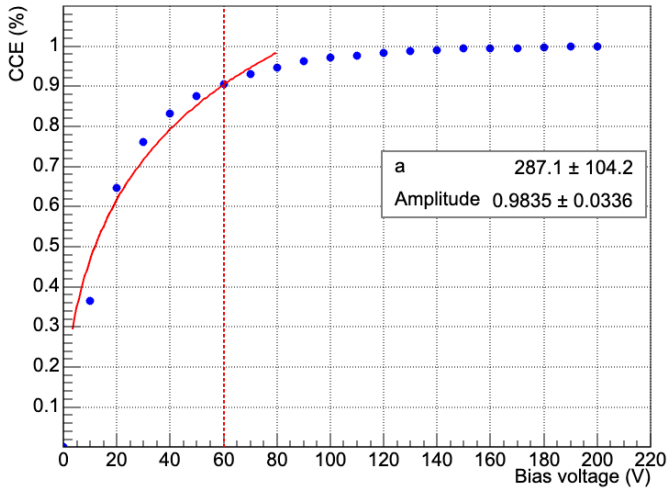


Figure 13: Charge Collection Efficiency (CCE) curve obtained with the laser. The dashed line indicates the estimated depletion voltage, and the red line represents the theoretical fit in the partially depleted region.

Theoretically, the CCE of the detector when irradiated by the built-in laser follows this exponential model:

$$\mathrm{CCE}(U) = \frac{1 - \exp\left(\frac{-d_c(U)}{a}\right)}{1 - \exp\left(\frac{-D}{a}\right)} \tag{5}$$

where D is the total sensor thickness ($D = 300\,\mu\mathrm{m}$) and $d_c(U)$ is the width of the depletion zone. In the partially depleted regime, where the applied voltage is lower than the depletion voltage ($U < U_{dep}$), the width of the depletion zone can be calculated as:

$$d_c(U) = D\sqrt{\frac{U}{U_{dep}}} \tag{6}$$

where U_{dep} is the depletion voltage, which was previously estimated to be 60 V for this measurement. By substituting Equation 6 into Equation 5, we performed a fit on the experimental data points in the non-plateau region to extract the penetration depth a of the laser into the silicon bulk. The fit, shown as a red line in Figure 13, yielded a penetration depth of $a = (287 \pm 104)\,\mu\mathrm{m}$.

3 Conclusions

The main objective of this experiment was to understand the operational principles of a silicon strip particle detector, a technology that plays a pivotal role in high-precision tracking systems like the ATLAS Semiconductor Tracker (SCT) at the LHC. By characterizing the electrical and optical responses of a p-in-n silicon sensor, this laboratory exercise successfully bridged the theoretical physics of semiconductors with the practical requirements of modern high-energy physics experiments.

The initial electrical characterization validated the necessary conditions for optimal particle detection. The Current-Voltage (I-V) measurements determined a depletion voltage of approximately 60 V, confirming the value required to fully deplete the 300 μm silicon bulk. Afterwards, the pedestal and noise analyses demonstrated that the system exhibits exceptionally low and stable intrinsic noise (averaging 1.9 ADC counts RMS) with no systematic electronic biases. Characterizing this noise is critical for large-scale experiments, as it ensures a high signal-to-noise ratio necessary for tracking minimum ionizing particles. Furthermore, the successful charge calibration established the essential transfer function required to translate raw ADC digital signals into physical deposited energy values.

Optical investigations using an infrared laser further verified the spatial capabilities of the sensor. The measured strip pitch of 160.0 μm aligns perfectly with the geometric design of the device, highlighting its high spatial resolution. Finally, the Charge Collection Efficiency (CCE) measurement allowed us to model the growth of the active volume. By fitting the efficiency curve in the partially depleted region, the penetration depth of the laser was estimated at $a = (287 \pm 104)\,\mu$m. This value is highly consistent with the 300 μm physical thickness of the sensor. While the parameter still exhibits a notable uncertainty—likely due to the limited number of data points in the narrow $0 - 60$ V fitting interval and extreme sensitivity to the optical focus of the laser spot—the measurement effectively validates the theoretical models of semiconductor depletion.

Ultimately, the experiment demonstrated that the characterized sensor behaves in excellent agreement with semiconductor theory, highlighting exactly why such robust, low-noise, and high-resolution devices are indispensable for tracking charged particles in the extreme collision environments of the LHC.

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